NCERT Class 10 Physics – Chapter 10: Light – Reflection and Refraction

CHAPTER 10: LIGHT – REFLECTION AND REFRACTION

10.1 LAWS OF REFLECTION

1. The angle of incidence is equal to the angle of reflection.
2. The incident ray, the normal to the mirror at the point of incidence and the reflected ray, all lie in the same plane.

These laws of reflection are applicable to all types of reflecting surfaces including spherical surfaces.

10.2 MIRROR FORMULA AND MAGNIFICATION

In a spherical mirror, the distance of the object from its pole is called the object distance (u). The distance of the image from the pole of the mirror is called the image distance (v). The distance of the principal focus from the pole is called the focal length (f). There is a relationship between these three quantities given by the mirror formula: $1 / v + 1 / u = 1 / f$. Magnification produced by a spherical mirror gives the relative extent to which the image of an object is magnified: m = Height of image (h') / Height of object (h) = $- v / u$.

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10.3 REFRACTION OF LIGHT AND SNELL'S LAW

Light does not travel in the same direction in all media. It appears that when travelling obliquely from one medium to another, the direction of propagation of light in the second medium changes. This phenomenon is known as refraction of light.

LAWS OF REFRACTION:

1. The incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane.
2. The ratio of sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given colour and for the given pair of media. This law is also known as Snell's law of refraction: sin i / sin r = constant = n21, where n21 is called the refractive index of the second medium with respect to the first medium.